

MDS-UPDRS Finger Tapping Assessment

The data provided below consists of extracted features (tap count, periods, and amplitudes) derived from a subject performing the MDS-UPDRS Item 3.4 Finger Tapping test. The subject was instructed to tap the index finger on the thumb 10 times as quickly and as largely as possible.

The objective is to score the subject according to MDS-UPDRS item 3.4: FINGER TAPPING based on these 10 consecutive finger taps. The testing instructions and scoring criteria for MDS-UPDRS item 3.4: FINGER TAPPING are as follows:

Instructions to the examiner:

Each hand is tested separately. Demonstrate the task, but do not continue to perform the task while the patient is being tested.

Instruct the patient to tap the index finger on the thumb 10 times as quickly AND as large as possible. Rate each side separately, evaluating speed, amplitude, hesitations, halts, and decrementing amplitude.

Scoring Criteria:

0 (Normal): No problems

1 (Slight): Any of the following:

- a) The regular rhythm is broken with one or two interruptions or hesitations of the tapping movement
- b) Slight slowing
- c) The amplitude decrements near the end of the 10 taps

2 (Mild): Any of the following:

- a) 3 to 5 interruptions during tapping
- b) Mild slowing
- c) The amplitude decrements midway through the 10-tap sequence

3 (Moderate): Any of the following:

- a) More than 5 interruptions during tapping or at least one longer arrest (freeze) in ongoing movement
- b) Moderate slowing
- c) The amplitude decrements starting after the 1st tap

4 (Severe): Cannot or can only barely perform the task due to slowing, interruptions, or decrements

Below are the extracted features including tap count, periods, and amplitudes. Please analyze this information and provide your score on the last line.

EXTRACTED FEATURES

Number of taps: 11

Tap periods (frames): [145, 156, 125, 170, 145, 154, 176, 142, 173, 150]

Tap amplitudes (pixels): [116.45600027478189, 119.9208072020865, 119.9208072020865, 127.12198865656563, 125.25174649480941, 107.2240644631605, 109.20164833920778, 110.30865786510141, 114.16216536138407, 109.78615577567146, 103.16006979447037]

GPT-5

Analysis of extracted features for MDS-UPDRS 3.4 Finger Tapping

- Task target: 10 taps; subject produced 11 detected peaks. We will evaluate the first 10 intervals (provided) and the amplitude trend over the full sequence to assess decrement.
- - Periods (proxy for speed and rhythm consistency), in frames:
[145, 156, 125, 170, 145, 154, 176, 142, 173, 150]
 - Mean period ≈ 153.6 frames; range 125–176 frames; spread $\approx \pm 15\%$ around the mean.
 - No obvious prolonged arrest or freeze; intervals fluctuate but remain within a moderate variability band.
 - Rhythm: there are multiple irregularities (e.g., short 125 followed by long 170; several long intervals ≥ 170). These suggest more than just 1–2 hesitations; likely 3+ rhythm interruptions across the 10 taps.
- - Amplitudes (pixels), 11 values:
[116.46, 119.92, 119.92, 127.12, 125.25, 107.22, 109.20, 110.31, 114.16, 109.79, 103.16]
 - Early taps increase to a peak at tap 4, then show a general downward trend from tap 5 onward.
 - Decrement pattern: amplitude begins to decline starting at tap 5 (i.e., after the first few taps), and continues with some small fluctuations, reaching the lowest value at the end (tap 11).

- This pattern is consistent with a decrement beginning midway through the sequence rather than only near the end or immediately after the first tap.

Mapping to scoring criteria:

- Interruptions/hesitations: The variability in periods suggests more than 1–2 hesitations, plausibly in the 3–5 range → aligns with score 2 (Mild).
- Speed: Period variability and overall mean do not indicate severe slowing; at most mild to moderate. Given the absence of very prolonged intervals or freezes, this supports score 1–2.
- Amplitude decrement: Begins around the middle of the sequence (after the 4th tap), not just near the end and not immediately after the first tap → matches score 2 (Mild).

Overall, the amplitude pattern (mid-sequence decrement) is the most specific discriminator and aligns with “2 (Mild).”

Score: 2
